joining the two charges. $\hat{r}_{12}$ is the unit vector pointing from charge q_1 to q_2 .It is shown in the Figure 1.2. Likewise, the force on the charge q_1 exerted by q_2 is along $-\hat{r}_{12}$ (i.e., in the direction opposite to $\hat{r}_{12}$).

(iii) In SI units, $k = \frac{1}{4\pi\epsilon_o}$ and its value is $9\times10^9\,\text{N m}^2\,\text{C}^{-2}$. Here ϵ_o is the permittivity of free space or vacuum and its value is

$$\epsilon_o = \frac{1}{4\pi k} = 8.85\times10^{-12}\,\text{C}^2\text{N}^{-1}\text{m}^{-2}.$$

(iv) The magnitude of the electrostatic force between two charges each of one coulomb and separated by a distance of 1 m is calculated as follows:

$$|F| = \frac{9\times10^9\times1\times1}{1^2} = 9\times10^9\,\text{N}.$$

This is a huge quantity, almost equivalent to the weight of one million ton. We never come across 1 coulomb of charge in practice. Most of the electrical phenomena in day-to-day life involve electrical charges of the order of μC (micro coulomb) or nC (nano coulomb).

(v) In SI units, Coulomb's law in vacuum takes the form $\vec{F}_{21} = \frac{1}{4\pi\epsilon_\circ}\frac{q_1q_2}{r^2}\hat{r}_{12}$. In a medium of permittivity ϵ, the force between two point charges is given by $\vec{F}_{21} = \frac{1}{4\pi\epsilon}\frac{q_1q_2}{r^2}\hat{r}_{12}$. Since $\epsilon>\epsilon_o$, the force between two point charges in a medium other than vacuum is always less than that in vacuum. We define the relative permittivity for a given medium as $\epsilon_r = \frac{\epsilon}{\epsilon_\circ}$. For vacuum or air, $\epsilon_r = 1$ and for all other media $\epsilon_r > 1$.

(vi) Coulomb's law has same structure as Newton's law of gravitation. Both are inversely proportional to the square of the distance between the particles. The electrostatic force is directly proportional to the product of the magnitude of two point charges and gravitational force is directly proportional to the product of two masses. But there are some important differences between these two laws.

- The gravitational force between two masses is always attractive but Coulomb force between two charges can be attractive or repulsive, depending on the nature of charges.
- The value of the gravitational constant $G = 6.67 \times 10^{-11}\ \text{N m}^2\,\text{kg}^{-2}$. The value of the constant k in Coulomb law is $k = 9 \times 10^9\ \text{N m}^2\,\text{C}^{-2}$. Since k is much more greater than G, the electrostatic force is always greater in magnitude than gravitational force for smaller size objects.
- The gravitational force between two masses is independent of the medium. For example, if 1 kg of two masses are kept in air or inside water, the gravitational force between two masses remains the same. But the electrostatic force between the two charges depends on nature of the medium in which the two charges are kept at rest.

(vii) The force on a charge q_1 exerted by a point charge q_2 is given by

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_\circ}\frac{q_1q_2}{r^2}\hat{r}_{21}$$

Here $\hat{r}_{21}$ is the unit vector from charge q_2 to q_1.

But $\hat{r}_{21} = -\hat{r}_{12}$,

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